

Dewell Guerand

Belgium | guerand.com | github.com/DewellGuerand | linkedin.com/GuerandDewell

Engineering student specialising in Applied Mathematics, passionate about applying the tools learned during my studies to real-world problems. I love having an impact when I do something and seeing tangible results. I have a strong interest in Machine Learning, Finite Element Methods, and Stochastic Differential Equations.

Education

MEng Double Degree , CentraleSupélec, Université Paris-Saclay	Aug 2026 – June 2028
• Engineering cycle – T.I.M.E (Top International Managers in Engineering) • Ranked 2nd in Mathematics worldwide by the ShangaiRanking	
MEng , Louvain School of Engineering (EPL), UCLouvain	Sept 2025 – June 2027
• Major: Applied Mathematics	
BEng , Louvain School of Engineering (EPL), UCLouvain	Sept 2022 – June 2025
• Major: Applied Mathematics, Minor: Computer Science • Graduated <i>Cum Laude</i>	
Lycée de Berlaymont , Belgium	Sept 2016 – June 2022
• Graduated with honors – Major in Mathematics and Sciences	

Experience

Quant Analyst , LSM Investment Club – UCLouvain	Oct 2025 – June 2026
• Conducted quantitative analysis and financial modelling to support investment decisions across equity. • As the only applied math majors of the club, I presented and popularized recent academic research in quantitative finance to club members (Markov chain, option and structuring, coordination of IA agents).	

Projects

Interest Rate Modeling – Sinusoidal Hull-White Model	github.com/DewellGuerand
• Implemented and validated an interest rate simulation engine based on the Hull-White model, incorporating a sinusoidal mean-reversion extension derived from a research paper; identified inconsistencies in the paper's derivations, calibrated model parameters, and benchmarked the extension against standard short-rate models to assess the trade-offs of each approach.	
Pricer for GMAB & GMDB Insurance Products	
• As part of my course on Financial Valuation of Actuarial Liabilities, developed a full pricing framework for Guaranteed Minimum Accumulation Benefit (GMAB) and Guaranteed Minimum Death Benefit (GMDB) contracts, covering closed-form analytical derivations and numerical methods including Binomial Trees and Monte Carlo simulation (Euler-Maruyama scheme).	
Disease Detection – Machine Learning	
• Designed and trained a disease classification pipeline to support early diagnosis from medical imaging (Machine learning) combining a CNN for image-based feature extraction with ensemble methods (XGBoost, RandomForest); performed hyperparameter tuning and cross-validation, achieving a test RMSE of 0.0372 (i.e., an average prediction error of under 4%).	
Finite Element Analysis – 2D Linear Elasticity	
• In order to measure and compare the structural resistance of cardboard using triangular and hexagonal mesh patterns, developed a complete finite element analysis application for solving 2D linear elasticity problems in engineering structures, with mesh generation, solver, and visualization components.	

Personal Achievements

- **Hockey:** 15+ years of competitive play at club level; coached and mentored junior players.
- **Running:** Completed the Brussels 20 km (half-marathon distance), training consistently while managing a full academic workload.
- **Senior Prom Trek:** Ranked 13th out of 120 High schools in the Walloon region in a 40 km endurance trek.

Skills & Languages

Programming: Python, C, C++, Java, Bash, Oz

Libraries: NumPy, SciPy, Matplotlib, Pandas

Tools: Git, Overleaf, Typst

Languages: French (C2, native) | English (C1, IELTS 7.0) | Dutch (B2)